



Radio Astronomy Lecture 1

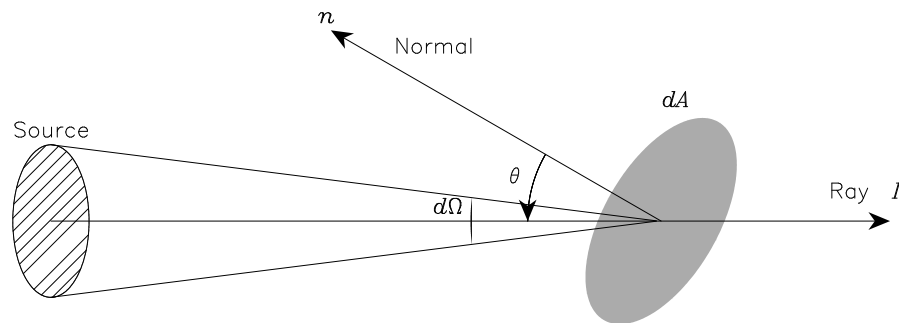
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Specific Intensity or Brightness I_ν

Once the wavelength $\ll$ the scale of the system, radiation travels in straight lines called Rays. The energy of a source dE crossing a surface dA in time dt and in frequency $d\nu$ is



$$dE = I_\nu \cos \theta dA dt d\Omega d\nu$$

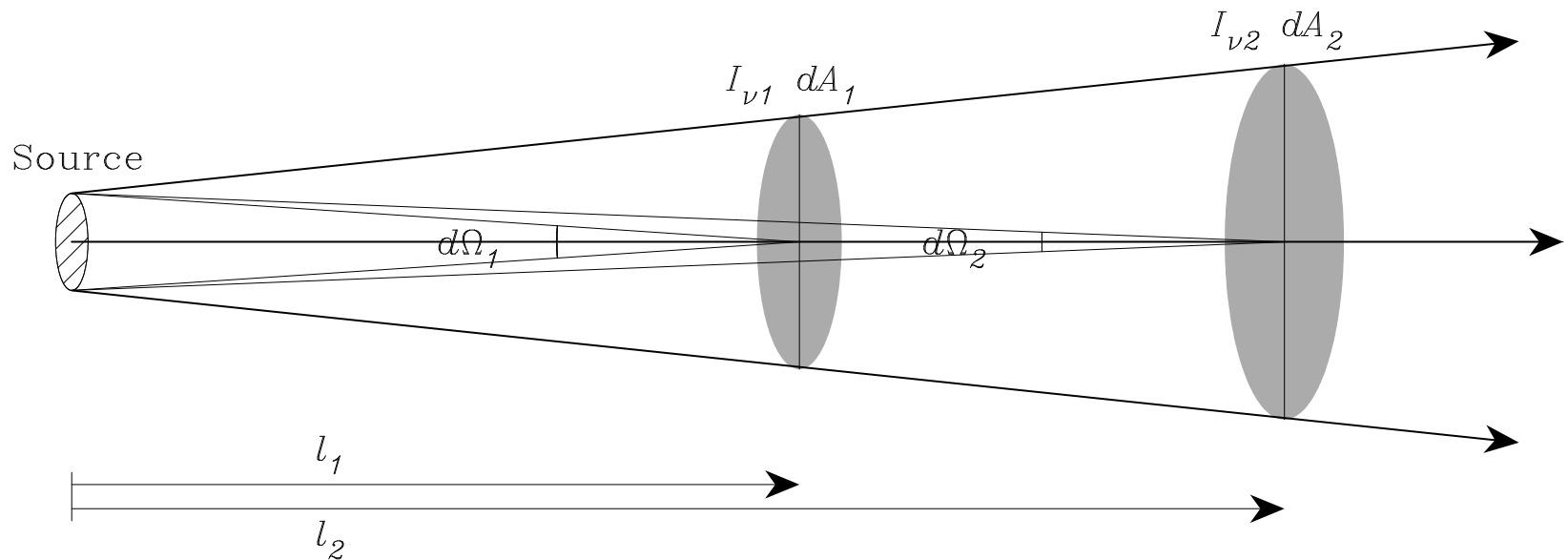
with $I_\nu(\theta, \phi) =$ specific intensity or brightness or energy flux ($\text{erg s}^{-1} \text{ cm}^{-2} \text{ sr}^{-1} \text{ Hz}^{-1}$)

- $d\Omega = \sin \theta d\theta d\phi$: solid angle (sr^{-1}) **subtended by the source!!**
- θ : angle betw. the normal and the Ray's direction
- dA : **area of the receiving surface**

Please note how $d\Omega$ and dA are measured. What if $\theta = 0$?

Constancy of Specific Intensity

In free space, the specific intensity is constant along the path.



$$dE = I_{\nu 1} dA_1 dt d\Omega_1 d\nu = I_{\nu 2} dA_2 dt d\Omega_2 d\nu$$

Since $dA_1/dA_2 = (l_1/l_2)^2$ and $d\Omega_1/d\Omega_2 = (l_2/l_1)^2$, we have

$$I_{\nu 1} = I_{\nu 2}$$

Radiation Energy Density u_ν

Radiation travels at the speed of light c and the **energy flux** carried by the radiation is I_ν . Therefore, the radiation energy density, which is defined as the energy per unit volume per unit frequency per steradian, is simply

$$u_\nu = \frac{I_\nu}{c}$$

in unit of $\text{erg cm}^{-3} \text{ sr}^{-1} \text{ Hz}^{-1}$. Or

$$I_\nu = cu_\nu$$

Flux or Flux Density F_ν

The flux passing through a surface (such as a detector) is given by

$$F_\nu = \int I_\nu(\theta, \phi) \cos \theta d\Omega$$

with a unit of

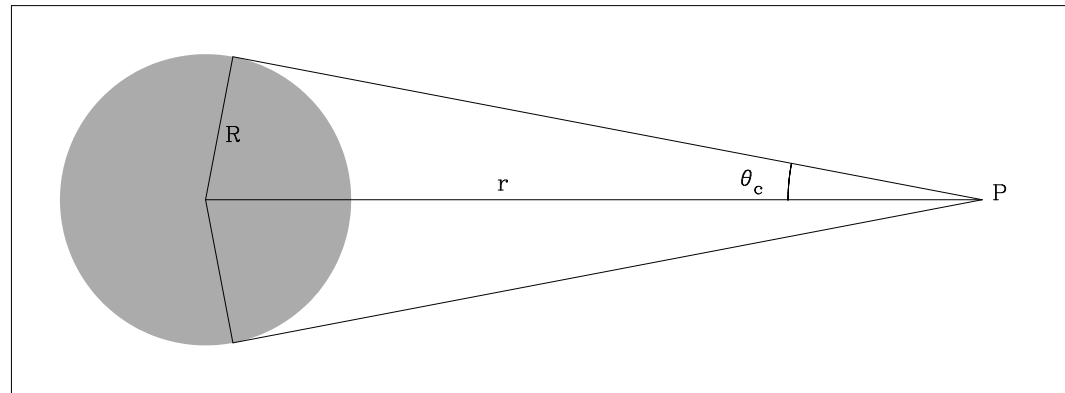
$$\text{erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$$

Since the flux for most of the radio sources is pretty small, we define a unit of Jansky (Jy) (after Karl Jansky) with

$$1 \text{ Jy} = 10^{-23} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ Hz}^{-1}$$

Total Flux Received from a Source

Consider a sphere with uniform brightness I_ν with a radius R . The total flux received by an observer at the distance r is:



$$F_\nu = \int_{\Omega_s} I_\nu(\theta, \phi) \cos \theta d\Omega = I_\nu \int_0^{2\pi} d\phi \int_0^{\theta_c} \sin \theta \cos \theta d\theta$$

$$\sin \theta_c = \frac{R}{r}$$

Thus,

$$F_\nu = \pi I_\nu \sin^2 \theta_c = I_\nu \frac{\pi R^2}{r^2} \rightarrow I_\nu \Delta\Omega \text{ as } r \gg R \text{ with } \Delta\Omega = \frac{\pi R^2}{r^2}$$

Radiative Transfer

As a ray passes through matter, energy may be added or subtracted from it by emission or absorption, and I_ν will not remain constant. Ignore scattering of photons at the moment. Thus,

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu$$

where

- s : distance along the path (cm)
- α_ν : absorption coefficient (cm^{-1})
- j_ν : emission coefficient ($\text{erg cm}^{-3} \text{ s}^{-1}$) $\text{sr}^{-1} \text{ Hz}^{-1} = \frac{P_\nu}{4\pi}$ if isotropic

Radiative Transfer - Special cases

- Emission Only: $\alpha_\nu = 0$, then

$$\frac{dI_\nu}{ds} = j_\nu$$

which has the sol'n

$$I_\nu(s) = I_\nu(s_o) + \int_{s_o}^s j_\nu(s') ds'$$

The increase in brightness equals the emission coeff. integrated along the path.

- Absorption Only: $j_\nu = 0$, then

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu$$

which has the sol'n

$$I_\nu(s) = I_\nu(s_o) \exp \left[- \int_{s_o}^s \alpha_\nu(s') ds' \right]$$

The brightness decreases exponentially.

Radiative Transfer - Optical depth

The transfer equation takes a simple form if, instead of s , we use another variable τ_ν called the optical depth, defined by

$$d\tau_\nu = \alpha_\nu ds$$

Then, the transfer equation

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu$$

can be rewritten as follows

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu$$

where the [source function](#) is defined as

$$S_\nu \equiv \frac{j_\nu}{\alpha_\nu}$$

Radiative Transfer - Formal soln

The formal soln of the transfer eq. is

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + \int_0^{\tau_\nu} e^{-(\tau_\nu - \tau'_\nu)} S_\nu(\tau'_\nu) d\tau'_\nu$$

It is a sum of (1) the initial intensity diminished by absorption plus (2) the integrated source diminished by absorption. i.e.,

- At $\tau'_\nu = 0$: integral in (2) becomes $S_\nu(0)e^{-\tau_\nu}$
- At $\tau'_\nu = \tau_\nu$: integral in (2) becomes $S_\nu(\tau_\nu)$

If S_ν is constant, we have

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu})$$

Blackbody Radiation

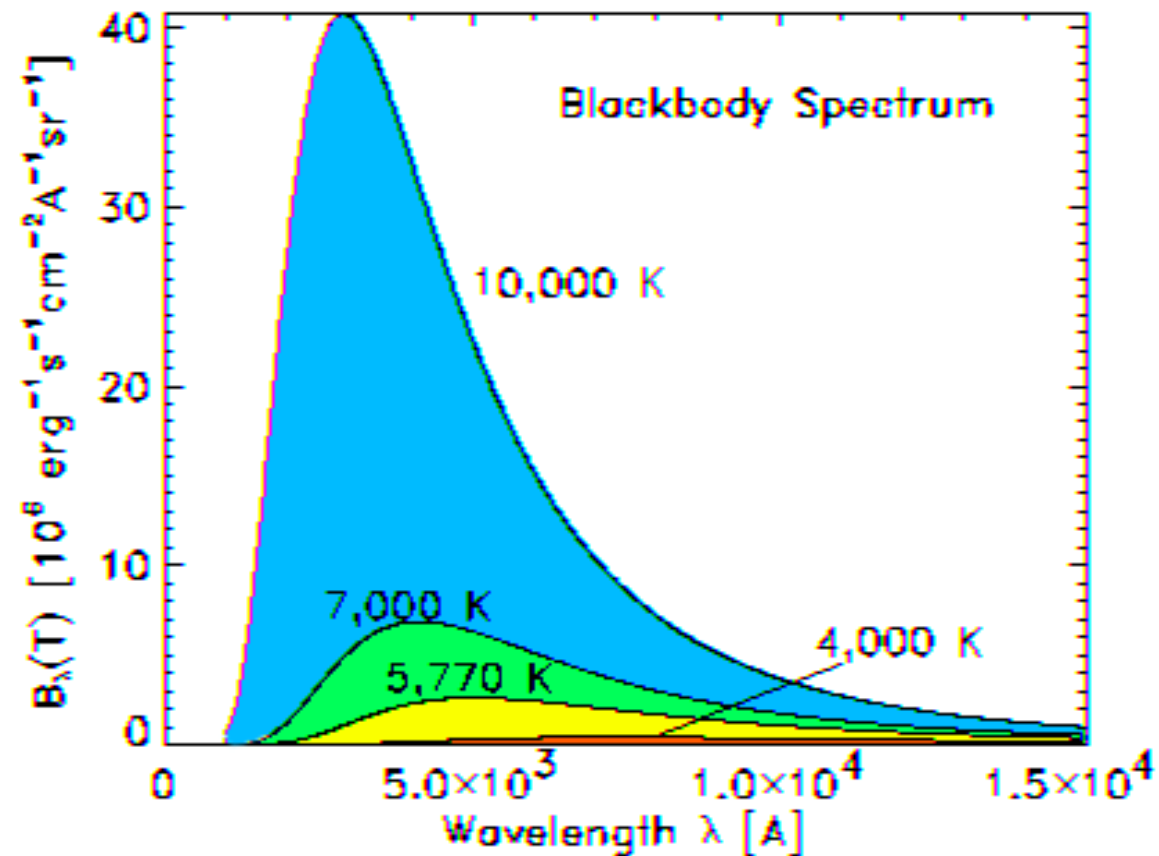
A **black body** is an object that absorbs all EM radiation that falls on it. No EM radiation passes through it and none is reflected. Because no light (visible EM radiation) is reflected or transmitted, the object appears black when it is cold.

If a perfect black body at a certain temperature is surrounded by other objects in **thermal equilibrium** at the same temperature, it will on average emit exactly as much as it absorbs, at every wavelength.

The radiation emitted by a black body is called **black-body radiation**.

In the lab., black-body radiation is approximated by the radiation from a small hole entrance to a large cavity (i.e, **cavity radiation**). Any light entering the hole would have to reflect off the walls of the cavity multiple times before it escaped, in which process it is nearly certain to be absorbed. This occurs regardless of the wavelength of the radiation entering (as long as it is small compared to the hole). The hole, then, is a close approximation of a theoretical black body and, if the cavity is heated, the spectrum of the hole's radiation will be continuous, and will not depend on the material in the cavity. By a theorem proved by Kirchhoff (thermal equilibrium), this curve depends only on T of the cavity walls.

Blackbody Radiation - Spectrum



Blackbody Radiation

The radiation emitted by a black body is called black-body radiation. A radiation which is itself in (full) thermal equilibrium with the surroundings, i.e., no net energy is transferred between the radiation and the surroundings, i.e.,

$$\frac{dI_\nu}{ds} = 0$$

In such a medium, we have

$$I_\nu = B_\nu(T)$$

with B_ν given by the Planck function:

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

where

- h : Planck's constant 6.63×10^{-27} erg s
- k : Boltzmann's constant 1.38×10^{-16} erg K
- c : speed of light 3×10^{10} cm s⁻¹

Thermal Radiation

Full thermal equilibrium will happen only in very special circumstances such as in a black enclosure or in very dense region like stellar interiors. Most matter is only in local thermal equilibrium (LTE). Thermal radiation is radiation emitted by matter in LTE. From Kirchhoff's law, LTE requires that

$$S_\nu(T) = B_\nu(T)$$

Thus,

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + B_\nu(T)$$

For an isothermal medium, T is constant, we have (see previous slide)

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + B_\nu(1 - e^{-\tau_\nu})$$

Thermal radiation becomes blackbody radiation only for optically thick media, with $\tau_\nu \gg 1$. The radiation will be **saturated at the blackbody radiation**. Adding more material will not increase the radiation.

Blackbody Radiation - More

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

Since $B_\nu d\nu = -B_\lambda d\lambda$ and $d\nu = (-c/\lambda^2)d\lambda$, we also have

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

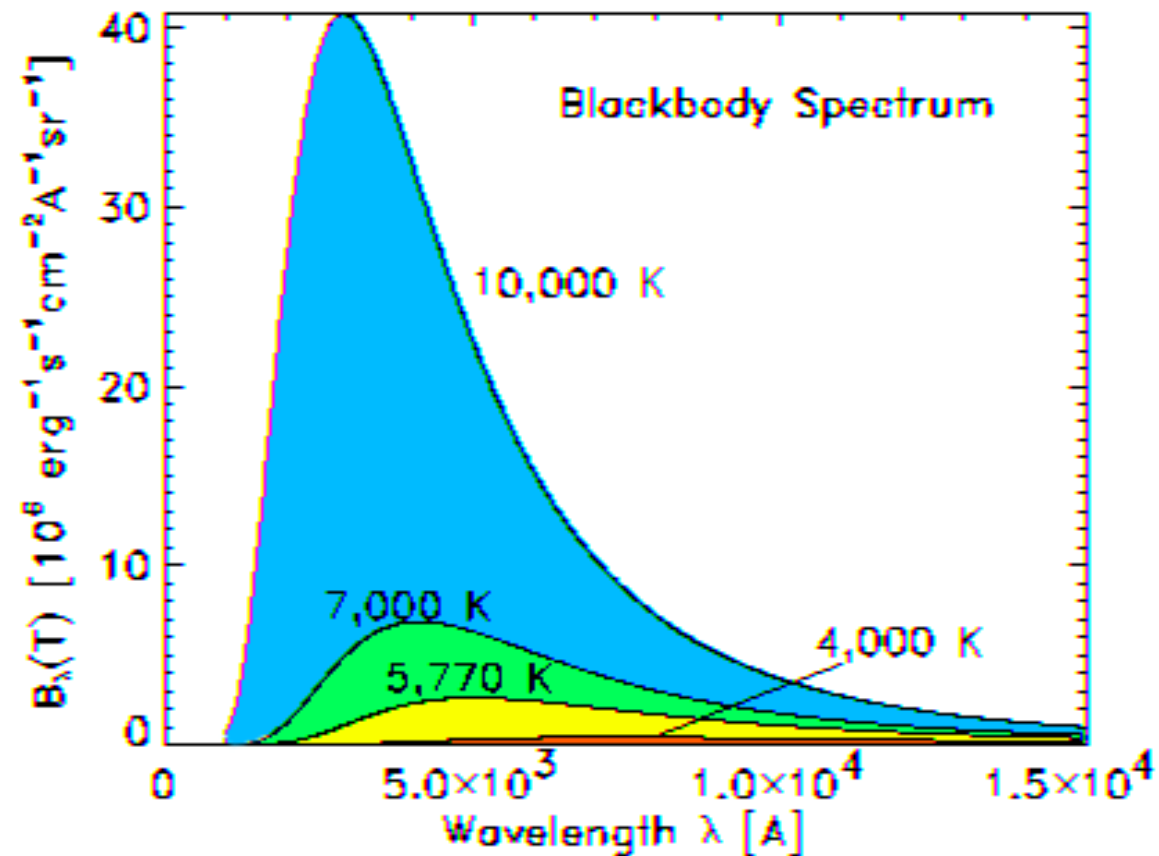
Total brightness of a blackbody radiation is

$$B(T) = \int_0^\infty B_\nu d\nu$$

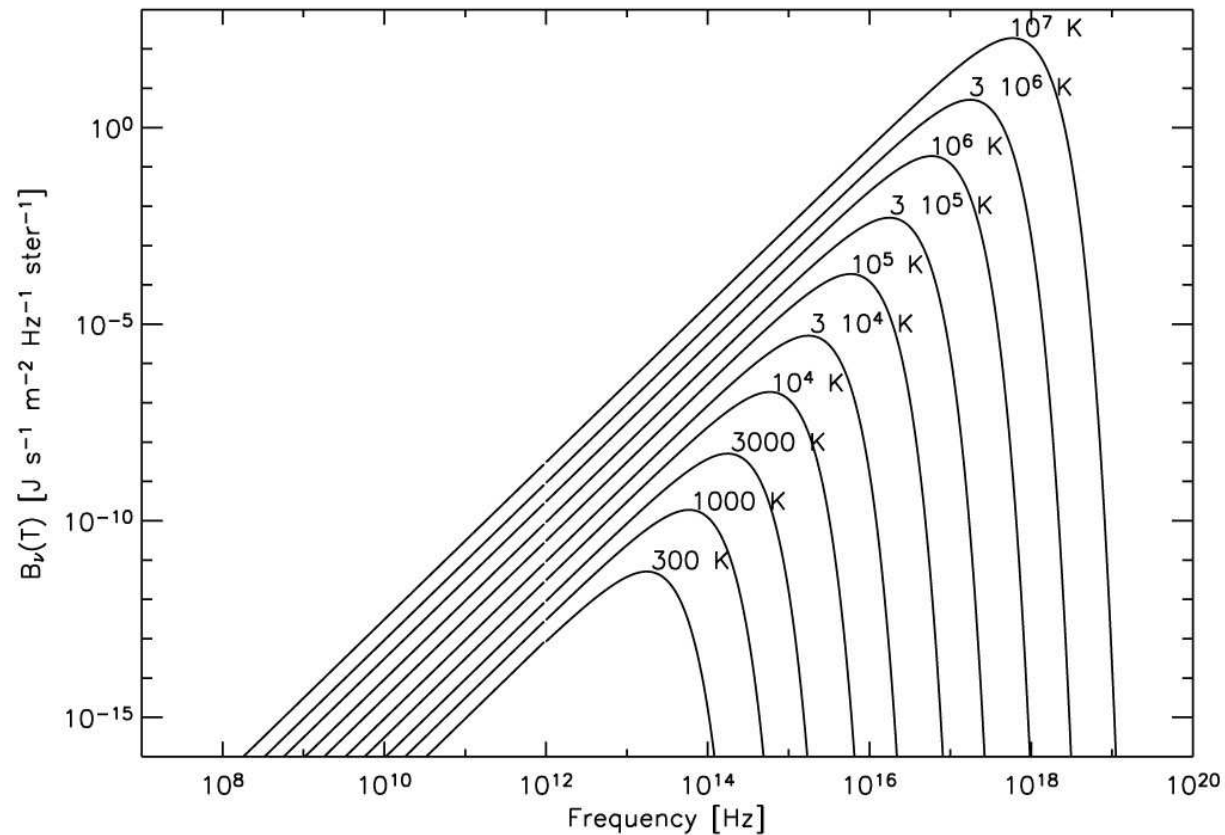
After some manipulation, we have (Stefan-Boltzmann radiation law)

$$B(T) = \sigma T^4, \quad \sigma = \frac{2\pi^4 k^4}{15c^2 h^3} = 1.8047 \times 10^{-5} \text{ erg cm}^{-2} \text{ s}^{-1} \text{ K}^{-4}$$

Blackbody Radiation - Spectrum



Blackbody Radiation - Spectrum



When T increase, the peak moves to higher ν and more energy at all ν .

Blackbody Radiation - Wien's Displacement

The frequency ν_{max} at which the peak of $B_\nu(T)$ occurs can be found by solving

$$\left. \frac{\partial B_\nu}{\partial \nu} \right|_{\nu=\nu_{max}} = 0$$

and

$$h\nu_{max} = 2.82kT$$

or

$$\frac{\nu_{max}}{T} = 5.88 \times 10^{10} \text{ Hz K}^{-1}$$

Similarly,

$$\left. \frac{\partial B_\lambda}{\partial \lambda} \right|_{\lambda=\lambda_{max}} = 0$$

and

$$\lambda_{max}T = 0.290 \text{ cm K}$$

Blackbody Radiation - Properties

- $h\nu \ll kT$: The Rayleigh-Jeans Law. In this case, the exponential can be expanded

$$\exp\left(\frac{h\nu}{kT}\right) = 1 + \frac{h\nu}{kT} + ..$$

Dropping the higher order terms, we have

$$B_{\nu}^{RJ} = \frac{2\nu^2}{c^2} kT$$

- $h\nu \gg kT$: Wien Law. In this case, $\exp(h\nu/kT) \gg 1$, and thus

$$B_{\nu}^W = \frac{2h\nu^3}{c^2} \exp\left(-\frac{h\nu}{kT}\right)$$

Brightness Temperature

In radio astronomy, we define a Brightness Temperature as

$$T_b = \frac{c^2}{2k\nu^2} I_\nu$$

Why this definition? If the emission is thermal (i.e., $S_\nu = B_\nu$) and the observed frequency is in the Rayleigh-Jeans limit, which is mostly true in the radio regime, then

$$T_b = \frac{c^2}{2k\nu^2} B_\nu = \frac{c^2}{2k\nu^2} \left(\frac{2\nu^2}{c^2} kT \right) = T$$

where T is the temperature of the material. So, the radiative transfer can be given by

$$\frac{dT_b}{d\tau_\nu} = -T_b + T$$

When T is constant,

$$T_b = T_b(0)e^{-\tau_\nu} + T(1 - e^{-\tau_\nu}), \quad h\nu \ll kT$$

- for optically thick with $\tau_\nu \gg 1$: $T_b = T$
- for optically thin with $\tau_\nu \ll 1$: $T_b = \tau_\nu T$